

UNIT -1

2. File System vs DBMS

File System

1. File system is a method of storing data in files.
2. Data is stored in operating system files.
3. Each application maintains its own files.
4. No centralized control over data.
5. Data is stored in flat files (like .txt, .dat).
6. Data redundancy is high.
7. Same data may be repeated in many files.
8. Data inconsistency occurs easily.
9. Updating data is difficult.
10. No data sharing between applications.
11. Programs are dependent on file structure.
12. Any change in file structure affects program.
13. Security is very low.
14. Backup and recovery is not automatic.
15. Searching data is slow.
16. No query language is used.
17. No data relationships are maintained.
18. Difficult to handle large datasets.
19. No data integrity rules.
20. Used in simple systems only.

◆ **DBMS (Database Management System)**

21. DBMS is software to manage databases.
22. Data is stored in organized format (tables).

23. Centralized control of data exists.
24. Multiple users can access data.
25. Data redundancy is minimized.
26. Data consistency is maintained.
27. Data sharing is easy.
28. Data is secure with access control.
29. Uses query language like SQL.
30. Faster data retrieval.
31. Supports data integrity constraints.
32. Backup and recovery are automatic.
33. Reduces program-data dependency.
34. Data abstraction is supported.
35. Provides different levels of data view.
36. Complex data handling is easy.
37. Supports relationships between data.
38. Transaction management is supported.
39. Concurrency control is available.
40. Suitable for large systems.

Key Differences (Comparison)

41. File system stores data in files; DBMS stores data in tables.
42. File system has no central control; DBMS has centralized control.
43. File system has high redundancy; DBMS reduces redundancy.
44. File system has low security; DBMS provides high security.
45. File system has data inconsistency; DBMS maintains consistency.
46. File system is slow in search; DBMS is fast.
47. File system has no query language; DBMS uses SQL.

48. File system has no data independence; DBMS supports it.

49. File system is difficult to maintain; DBMS is easy to maintain.

50. DBMS is more reliable and efficient than file system.

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